DUKE AIPI 590 · ALTERNATIVE DATA · MID-SEMESTER UPDATE

Climate Impacts on Futures Pricing

The physical-risk model works. The trading signal did not.



Haydn Stucker · October 14, 2026

Project overview

One project, two different tests



MEASURE

Can public climate data identify real crop supply shocks?

Compare weather and drought features with held-out crop yields.



PREDICT

Can those signals predict tradeable commodity returns?

Compare climate models with market-only and simple-rule baselines.

Intended impact: better crop-risk measurement and an honest test of whether public data adds market timing value.

Progress since the proposal

The project moved from exploratory correlations to execution-grade tests

1

DATA PIPELINE

Automated public climate, drought, yield and futures data

2

PHYSICAL MODEL

County weather and drought validated against held-out corn yields

3

MARKET TESTS

Dated LSEG contracts, costs, rolls and point-in-time labels

4

STRESS TESTS

Corn forecasts, USDA releases, cocoa and coffee

5

DECISION

No trading specification clears the final validation gate

Current status: the original trading hypothesis failed a much stronger test than the proposal could support.

Data collection and preparation

Every feature is aligned to when it became available

PUBLIC CLIMATE

NASA POWER, NOAA drought and GEFS forecasts

CROP OUTCOMES

County production, annual yield and USDA progress

MARKET DATA

Dated LSEG corn, wheat, cocoa and coffee contracts

POINT-IN-TIME PANEL

One decision row with the data actually available at that moment

Publication lag	Store release timestamps and prohibit future data
Missing and uneven coverage	Keep missing values explicit and impute only inside training
Contract rolls and trading costs	Use dated contracts, next-open entries and row-level costs

Sources: NASA POWER, NOAA, USDA NASS, US Drought Monitor and LSEG Workspace.

Model development

Climate models had to beat market-only and simple-rule baselines

BASELINE

Market-only model

Recent returns, volatility, curve shape and seasonality, with the same dates and costs as every climate candidate.

+ fixed trading rules

CANDIDATE FEATURES



Observed stress

Heat, rainfall and drought



Forecast revisions

Operational GEFS changes



Scheduled releases

USDA crop condition and progress

MODEL FAMILIES

Ridge regression

Logistic direction model

Shallow boosted trees

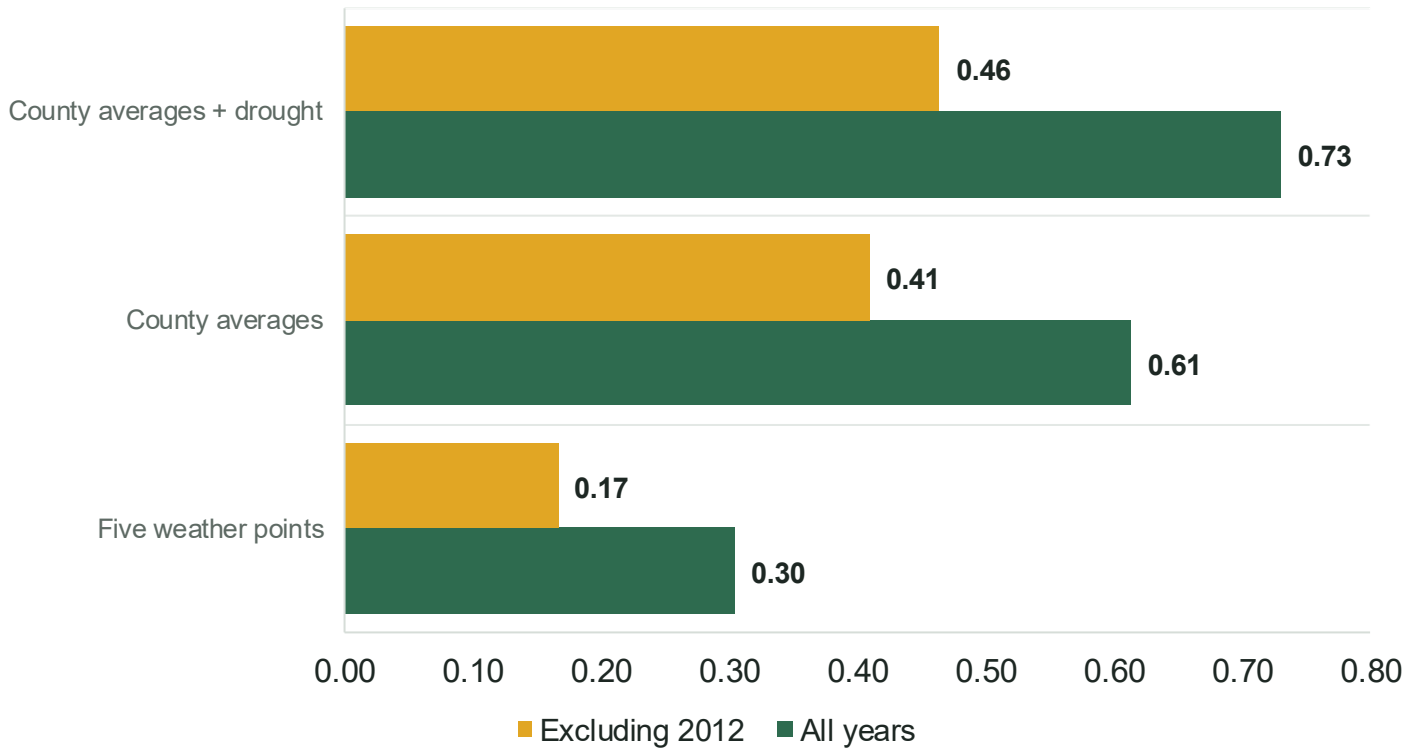
Selection ended before the final test period

All models use expanding time splits, target embargoes, non-overlapping trades and modeled execution costs.

What worked: measuring corn supply risk

County weather and drought predict held-out yield shortfalls

Held-out yield R²



0.73

Out-of-sample R² across 26 held-out harvests

0.46

R² after removing the exceptional 2012 drought

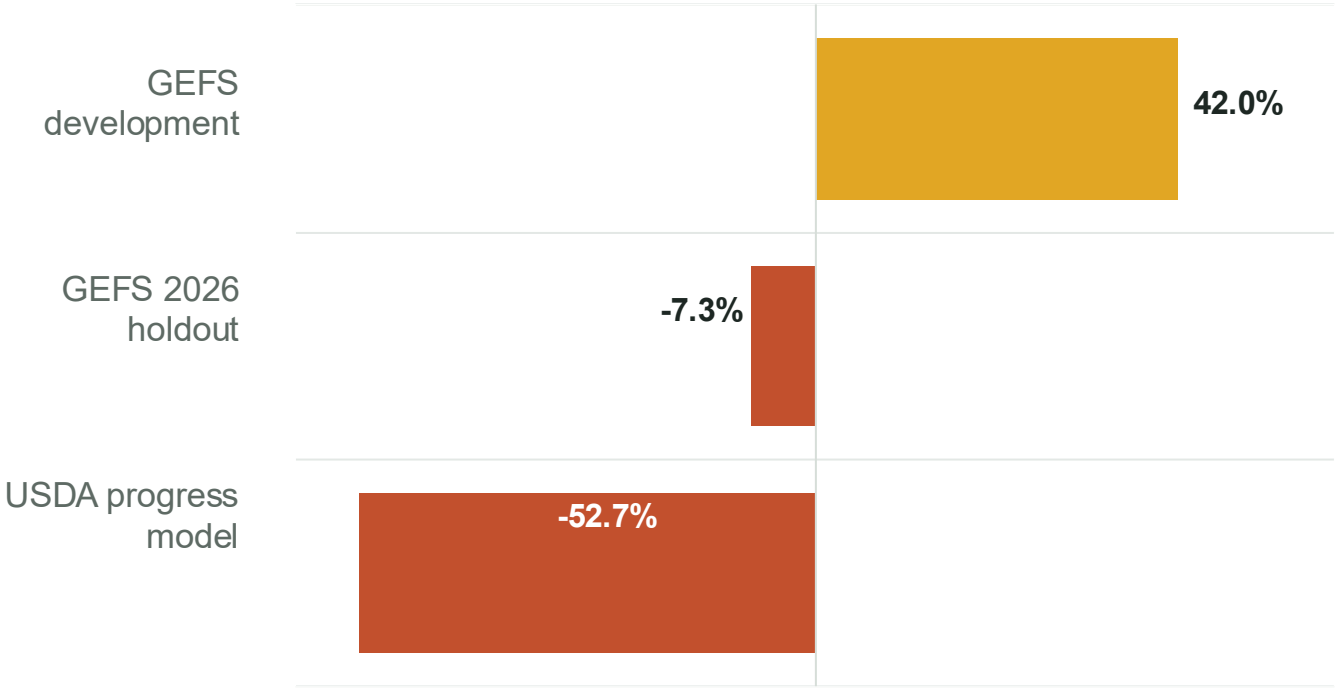
The physical measurement generalizes beyond one extreme year

Metric: leave-one-year-out R². Source: project corn-yield validation, 2000–2025.

What failed: repeatable buy and sell signals

Promising development results did not survive stricter validation

Net return after modeled costs



CURRENT DECISION

No buy or sell model is ready for use

−0.98	GEFS holdout Sharpe
3 / 9	Positive USDA years
0	Strategies promoted

Evaluation: net after costs, Sharpe, drawdown, positive years, doubled costs and matched baselines

Different samples are shown. Replication and gate outcomes matter more than cross-sample magnitude.

Challenges, risks and responses

The main risk is finding a story in noise

RISK	RESPONSE
Many model choices can create false confidence	Freeze candidates and gates before the final test
Extreme weather events are scarce and uneven	Use time-based validation and report null results
Public observations may arrive after prices move	Track publication times and test operational forecasts

A failed predeclared test closes a specification. It does not justify tuning against the failure.

NEXT STEPS

A supply-risk model is the defensible final deliverable

The final project will estimate crop supply risk from point-in-time climate data and document why the tested trading signals do not generalize.

1. **Calibrate yield-risk probabilities and uncertainty**
2. **Package the reproducible pipeline and model card**
3. **Write the final comparison and document the null result**

Questions?

3 minutes

github.com/haydn-s/climate-futures-pricing